Supplementary Material for "Robust High-Order Control Barrier Function Safety Filtering for Supercritical Power Plant Control under Model Mismatch"

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1. Supplementary Material

This supplement provides detailed experimental results and extended analyses referenced in the main text.

1.1. S1. Generic-GP and Out-of-Distribution Deployment

The main text (Section 7) reports that scenario-specific GP training achieves 0% CBF violation under nominal operation and the six static perturbation deployments. This supplement provides the full OOD analysis: Table S1 reports CBF violation and QP intervention rates under three GP training strategies—scenario-specific, generic mixed (trained on Nominal/S1/S2/S4), and mismatched OOD (S1→S3, S3→S1)—using Φ-scaled rollout, $\kappa_\epsilon = 1.0$, and 5 seeds × 500 evaluation steps.

Three observations follow. (1) Scenario-specific GP is uniformly safe but with near-saturated QP intervention under perturbation. (2) Generic mixed GP fails under all perturbed deployments (82.7–99.7% violation) because it averages over training scenarios, producing σ_{GP} too small to trigger the safety filter—a

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Table S1: Generic-GP and OOD deployment — CBF violation rate (%, mean $\pm$ std over 5 seeds $\times$ 500 steps).

Deployment	Strategy	CBF (%)	QP (%)
Nominal	scenario-specific	$\mathbf{0.0 \pm 0.0}$	0.2
	generic mixed	$\mathbf{0.0 \pm 0.1}$	43.0
S1: Heat	scenario-specific	$\mathbf{0.0 \pm 0.0}$	100.0
	generic mixed	92.2 ± 4.6	7.9
	OOD (S3→S1)	99.9 ± 0.1	99.8
S2: Pressure	scenario-specific	$\mathbf{0.0 \pm 0.0}$	99.8
	generic mixed	96.4 ± 0.6	3.4
S3: Coupled	scenario-specific	$\mathbf{0.0 \pm 0.0}$	99.2
	generic mixed	82.7 ± 3.9	18.6
	OOD (S1→S3)	$\mathbf{0.0 \pm 0.0}$	100.0
S4: Nonlinear	scenario-specific	$\mathbf{0.0 \pm 0.0}$	99.6
	generic mixed	98.2 ± 0.3	1.1
S5: Valve	scenario-specific	$\mathbf{0.0 \pm 0.0}$	99.6
	generic mixed	98.8 ± 0.3	0.2
S6: Fuel	scenario-specific	$\mathbf{0.0 \pm 0.0}$	99.9
	generic mixed	99.7 ± 0.2	0.2

silent calibration failure. (3) OOD asymmetry: S1→S3 achieves 0% violation (large σ_{GP} inflates $\epsilon(x)$) while S3→S1 reaches 99.9% violation (low σ_{GP} despite biased mean)—σ_{GP} measures spatial distance, not functional distance between perturbation structures.

1.2. S2. Perturbation Magnitude Sweep

Table S2 reports the perturbation magnitude sweep on S1:Heat (3 seeds per magnitude, Φ-scaled rollout, $\kappa_\epsilon = 1.0$). The QP intervention rate transitions smoothly from selective (3.5% at Mag10) to saturated (100% at Mag75), with 0% CBF violation through Mag75. At Mag100, the perturbation exceeds the GP's modeling capacity and violation reaches 99.8%.

Table S2: Perturbation magnitude sweep — CBF violation and QP intervention vs. enthalpy perturbation magnitude (S1:Heat structure, 3 seeds $\times$ 500 steps).

Magnitude	Δf_h (kJ/kg)	CBF%	QP%
Mag10	-10	0.0 ± 0.0	3.5
Moderate	-15	0.0 ± 0.0	9.8
Mag25	-25	0.0 ± 0.0	49.4
Mag50	-50	0.0 ± 0.0	99.9
Mag75	-75	0.0 ± 0.0	100.0
Mag100	-100	99.8 ± 0.0	100.0

1.3. S3. κ_ϵ Sensitivity Sweep

Table S3 reports the κ_ϵ sweep on S1:Heat and S3:Coupled under Φ-scaled nonlinear dynamics (3 seeds). All $\kappa_\epsilon \in [0.3, 1.0]$ achieve 0% CBF violation; Theorem 1 formally certifies only $\kappa_\epsilon = 1$. Reward improves monotonically with κ_ϵ and chatter shifts by < 1.0 percentage point.

1.4. S4. Training Mode Ablation

Table S4 compares three training modes under S1:Heat. All achieve 0% CBF violation at deployment; decoupled mode requires 12% less training time;

Table S3: κ_ϵ sensitivity sweep (3 seeds, mean $\pm$ std).

κ_ϵ	Scenario	CBF %	Pwr %	Reward	Chatter %
0.30	S1	0.00 ± 0.00	0.00	-10919.9 ± 7.6	99.67
	S3	0.00 ± 0.00	0.00	-8089.5 ± 9.7	97.00
0.50	S1	0.00 ± 0.00	0.00	-10909.0 ± 7.8	99.67
	S3	0.00 ± 0.00	0.00	-8079.2 ± 9.7	97.40
0.70	S1	0.00 ± 0.00	0.00	-10898.3 ± 7.9	99.67
	S3	0.00 ± 0.00	0.00	-8068.9 ± 9.8	97.53
1.00	S1	$\mathbf{0.00 \pm 0.00}$	0.00	-10882.6 ± 8.3	99.67
	S3	$\mathbf{0.00 \pm 0.00}$	0.00	-8053.8 ± 9.9	97.93

coupled-differentiable is computationally infeasible (>30 min/seed).

Table S4: Training Mode Ablation under S1: Heat (mean $\pm$ std over 3 seeds, 30 eval episodes $\times$ 300 steps).

Mode	CBF%	Reward	Train Time	Solve (ms)
Decoupled	0.0	-4576.2 ± 1.8	12.5 ± 0.7 min	$\sim$25
Coupled-nondiff.	0.0	-4580.1 ± 2.1	14.2 ± 0.8 min	$\sim$25
Coupled-diff.	0.0	-4582.4 ± 2.5	> 30 min (no conv.)	$\sim$25

1.5. S5. GP–CBF Structural Comparison

Table S5 (moved from Section 2) summarizes the key structural distinctions between GP-augmented safety methods.

1.6. S6. Triple-Integrator Validation ($m = 3$)

The compositional robustness margin $\epsilon(x)$ is derived for general relative degree m (Theorem 1 of the main paper), but the CCS constraints have $m \leq 2$. This appendix validates the $m \geq 3$ generalization on a triple integrator system ($n = 3$, $m = 3$): $\dot{x}_1 = x_2$, $\dot{x}_2 = x_3$, $\dot{x}_3 = u$, with a circular keep-out zone $h(x) = (x_1 - c)^2 - r^2$ of relative degree $m = 3$. We evaluate four

Table S5: GP-augmented safety methods compared on three structural axes.

Method	Structural axis: Uncertainty propagation depth	Per-constraint differentiation	PAC-Bayes guarantee type
Cheng et al. [1, 2]	Top-level CBF only	No (single margin)	Per-step CBF (m=1)
Hewing et al. [3]	MPC prediction horizon (rollout)	Partial (per chance constraint, rollout-aggregated)	Chance-constrained over horizon
Ours (Robust HOCBF)	Full HOCBF ψ-chain (all m levels)	Yes (per constraint, by relative degree)	Per-step forward invariance ($m \geq 1$)

perturbation types: additive damping ($\Delta f = -\zeta[0, 0, x_2]^\top$, $\zeta = 0.5$), periodic ($\Delta f = A\sin(\omega t)[0, 0, 1]^\top$), coupled (damping + periodic), and nonlinear state-dependent ($\Delta f = -\eta[0, 0, x_1 x_2]^\top$). Each method is evaluated over 50 episodes of 500 steps.

Sigma-chain validation. A key property of the recursive σ-chain is that uncertainty grows monotonically with the ψ-chain level: $\sigma_1 < \sigma_2 < \sigma_3$. Table S6 validates this across 120 sampled states and all four perturbation scenarios. The ordering $\sigma_1 < \sigma_2 < \sigma_3$ holds for 100% of sampled states, with growth ratios $\sigma_2/\sigma_1 \approx$ 2.6–2.9 and $\sigma_3/\sigma_2 \approx$ 2.0–2.2, consistent with the recursive structure of (Eq. 11 in the main paper).

Table S6: Sigma-chain ordering validation ($m = 3$): mean σ_i values and growth ratios across 120 sampled states

Scenario	$\bar{\sigma}_1$	$\bar{\sigma}_2$	$\bar{\sigma}_3$	$\sigma_1 < \sigma_2 < \sigma_3$	σ_3/σ_1
Damping	0.077	0.207	0.437	100%	5.7
Periodic	0.101	0.254	0.514	100%	5.1
Coupled	0.101	0.254	0.514	100%	5.1
Nonlinear	0.077	0.219	0.474	100%	6.1

Safety evaluation. Table S7 reports the safety results. The critical result is under the *nonlinear* state-dependent perturbation: HOCBF(m=3) suffers 13.3% actual constraint violation and 41.5% CBF condition violation, while RHOCBF(m=3) achieves 0% actual constraint violation with only 3.5% CBF condition violation. The robustness margin $\epsilon(x)$ reduces actual violation from 13.3% to 0%.

Table S7: Triple Integrator Validation ($m = 3$): Constraint Violation Rate (%) under four perturbation types (50 episodes $\times$ 500 steps). "Actual" = entering keep-out zone ($h < 0$); "CBF" = CBF condition violation ($\psi_m < 0$).

Scenario	Method	Actual Viol.	CBF Viol.
Damping	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Periodic	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Coupled	HOCBF(m=3)	0.00	0.00
	RHOCBF(m=3)	0.00	0.00
Nonlinear	HOCBF(m=3)	13.25	41.45
	RHOCBF(m=3)	**0.00**	3.50

The 3.5% CBF condition violation under the nonlinear scenario reflects that the large $\bar{\epsilon} = 20.1$ occasionally renders the QP infeasible, in which case the raw policy action may violate the CBF condition. Crucially, the system remains a safe distance from the actual constraint boundary, achieving 0% actual violation—the compositional $\epsilon(x)$ provides practical safety even when formal forward invariance cannot be guaranteed due to QP infeasibility.

Compositional vs. constant ϵ. The nonlinear scenario provides a setting where $\epsilon(x)$ varies substantially ($\mu_\epsilon = 0.93$, $\max_\epsilon = 3.39$, $\mathrm{std}/\mu = 0.70$). Table S8 reports results across five seeds. All ϵ configurations achieve 0% actual violation, but CBF condition violation rates differ: compositional $\epsilon(x)$ (3.30%) outperforms ϵ_0^{mean} (4.70%) and underperforms $\epsilon_0^{\max}$ (1.50%)—the compositional $\epsilon(x)$ balances conservatism by being tighter in well-observed states while maintaining protection at the boundary.

Sparse GP coverage and task performance. A position constraint $h(x) = x_{\text{lim}} - x_1$ with constant gradient $|\partial h/\partial x_1| = 1$ combined with sparse GP ($n = 50$

Table S8: Triple Integrator ϵ Ablation under Nonlinear Perturbation ($m = 3$, 5 seeds $\times$ 50 episodes $\times$ 500 steps). Mean ϵ is the total per-step robustness margin; $\epsilon(x)$ statistics ($\mu = 0.93$, max $= 3.39$, std $= 0.65$) are computed over sampled states. Results: mean $\pm$ std over 5 seeds.

ϵ Configuration	Actual Viol. (%)	CBF Viol. (%)	Mean ϵ
Compositional $\epsilon(x)$	0.00 ± 0.00	3.30 ± 0.24	21.6
ϵ_0^{mean}	0.00 ± 0.00	4.70 ± 0.24	0.93
ϵ_0^{max}	0.00 ± 0.00	1.50 ± 0.00	3.39
No ϵ ($\epsilon = 0$)	13.43 ± 0.12	41.23 ± 0.07	0

samples confined to $x_1 \in [-0.3, 0.5]$, constraint at $x_{\text{lim}} = 1.2$) creates $17\times$ $\epsilon(x)$ variation. Table S9 shows that compositional $\epsilon(x)$ enables task completion that constant ϵ_0 prevents: $\bar{x}_1 = 0.37$ (31% of distance to constraint) with 0% violation vs. $\bar{x}_1 = -0.38$ and -1.33 for constant margins. Without ϵ, the state reaches $\bar{x}_1 = 0.59$ but with 12.2% violation.

Table S9: Sparse GP Ablation: Position Constraint with Partial GP Coverage ($m = 3$, nonlinear perturbation, 5 seeds $\times$ 10 episodes $\times$ 200 steps). GP trained on n=50 samples in $x_1 \in [-0.3, 0.5]$; constraint at $x_{\text{lim}} = 1.2$. Results: mean $\pm$ std over 5 seeds.

ϵ Config.	Viol. (%)	CBF Viol. (%)	QP Infeas. (%)	$\bar{x}_1$	$x_1^{\max}$	Mean ϵ
Compositional $\epsilon(x)$	0.00	0.00	0.0	**0.37**	**0.74**	0.15
ϵ_0^{mean}	0.00	0.00	0.0	-0.38	0.00	1.20
ϵ_0^{max}	0.00	0.00	0.2	-1.33	0.00	2.72
No ϵ ($\epsilon = 0$)	12.2 ± 0.0	56.8 ± 0.7	0.0	0.59	1.35	0

1.7. S7. Proofs of Theoretical Results

1.8. Lemma S1: Residual Gradient Bound

Lemma 1 (Residual gradient bound)**.** *Let $\delta_i = \psi_i - \hat{\psi}_i^0$ with $\delta_0 \equiv 0$. Under Assumption* **??***(b,c) and smoothness of $\hat{f}$ on the compact operating region $\mathcal{X}_{\text{op}}$,*

$\|\nabla_x \delta_i(x)\| \le G_\delta^{(i)}$ *for all* $x \in \mathcal{X}_{\mathrm{op}}$*, with*

$$G_\delta^{(0)} = 0,$$
$$G_\delta^{(i)} = H_\psi^{(i-1)}\|\Delta\hat{f}\|_\infty + G_\psi^{(i-1)} G_f + (L_{\hat{f},i} + k_i)G_\delta^{(i-1)} + H_\delta^{(i-1)}\|\Delta\hat{f}\|_\infty + G_\delta^{(i-1)} G_f \tag{1}$$

where $H_\psi^{(i-1)} = \sup_x \|\nabla_x^2 \hat{\psi}_{i-1}^0\|_{\mathrm{op}}$*,* $G_\psi^{(i-1)} = \sup_x \|\nabla_x \hat{\psi}_{i-1}^0\|$*, and* G_f *is the gradient bound from Assumption* **??***(c).*

Proof. Differentiate δ_i term by term: $\delta_i = L_{\Delta\hat{f}}\hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i)\delta_{i-1} + L_{\Delta\hat{f}}\delta_{i-1}$.

(i) $\nabla_x(L_{\Delta\hat{f}}\hat{\psi}_{i-1}^0) = \nabla_x^2\hat{\psi}_{i-1}^0 \Delta\hat{f} + (\nabla_x \Delta\hat{f})^\top \nabla_x \hat{\psi}_{i-1}^0$. By Cauchy-Schwarz and Assumption **??**(c): $\|\nabla_x(L_{\Delta\hat{f}}\hat{\psi}_{i-1}^0)\| \le H_\psi^{(i-1)}\|\Delta\hat{f}\|_\infty + G_\psi^{(i-1)}G_f$.

(ii) $\nabla_x((L_{\hat{f}} + k_i)\delta_{i-1})$ contributes $(L_{\hat{f},i} + k_i)G_\delta^{(i-1)}$.

(iii) $\nabla_x(L_{\Delta\hat{f}}\delta_{i-1}) = \nabla_x^2\delta_{i-1}\Delta\hat{f} + (\nabla_x\Delta\hat{f})^\top\nabla_x\delta_{i-1}$, bounded by $H_\delta^{(i-1)}\|\Delta\hat{f}\|_\infty + G_\delta^{(i-1)}G_f$.

Summing yields (1). Under Assumption **??**(c), $G_f = L_{\hat{f}}\|\Delta\hat{f}\|_\infty = \mathcal{O}(\|\Delta\hat{f}\|)$. By induction, $G_\delta^{(i)} = \mathcal{O}(\|\Delta\hat{f}\|)$ at every level, so $|L_{\Delta\hat{f}}\delta_{i-1}| \le G_\delta^{(i-1)}\|\Delta\hat{f}\| = \mathcal{O}(\|\Delta\hat{f}\|^2) = \mathcal{O}(\rho^2)$. □

1.9. Full Proof of Theorem 1

Proof of Theorem 1. **Step 1 (GP calibration).** By Assumptions **??**–**??** and the PAC-Bayes concentration [?], $|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \le \beta \cdot \sigma_{\mathrm{GP},j}(x)$ for all $x \in \mathcal{X}$ holds with probability $\ge 1 - \delta/n$ per dimension. By union bound over $j = 1, \ldots, n$, this holds simultaneously with probability $\ge 1 - \delta$. Condition on this event.

Step 2 (Compositional bound). We show $|\delta_i(x)| \le \sigma_i(x)$ by induction on i.

- *Base case ($i = 1$):* $\delta_1 = \sum_j (\partial h/\partial x_j)\Delta\hat{f}_j$. By Step 1 and triangle inequality, $|\delta_1| \le \beta \sum_j |\partial h/\partial x_j| \sigma_{\mathrm{GP},j} = \sigma_1$.

- *Inductive step ($i \ge 2$):* Assume $|\delta_{i-1}| \le \sigma_{i-1}$. From (**??**): $|\delta_i| \le |L_{\Delta\hat{f}}\hat{\psi}_{i-1}^0| + (|L_{\hat{f}}|_{\mathrm{op}} + k_i)|\delta_{i-1}| + |L_{\Delta\hat{f}}\delta_{i-1}|$. Term 1 bounded identically to base case;

Term 2 bounded by induction hypothesis; Term 3 bounded using Lemma 1: $|L_{\Delta\hat{f}}\delta_{i-1}| \leq G_\delta^{(i-1)}\beta\|\sigma_{\mathrm{GP}}\|_2 = \sigma_{\mathrm{cross}}^{(i)}$. Summing: $|\delta_i| \leq \sigma_i$.

The cross term is retained at full weight. Under Assumption **??**(c), it is formally $\mathcal{O}(\rho^2)$.

Step 3 (Total perturbation). Combining the inductive bounds from Step 2 with the coupling-coefficient bound (2) yields $|\Delta(x,u)| \leq \sigma_m(x) + \sum_{j=1}^{m-1} c_j\sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) = \sigma_{\mathrm{total}}(x)$. For $\kappa_\epsilon = 1$, $\epsilon(x) = \sigma_{\mathrm{total}}(x)$, hence $|\Delta(x,u)| \leq \epsilon(x)$.

Step 4 (Forward invariance). For u^* satisfying $A_0u^* \leq b_0 - \epsilon$: $\psi_m^{\mathrm{true}}(x,u^*) = \hat{\psi}_m^0(x,u^*) + \Delta(x,u^*) \geq \epsilon(x) - |\Delta(x,u^*)| \geq 0$. By Theorem 2 of [**?**], the safe set $\mathcal{C}$ is forward invariant. □

1.10. Coupling Coefficient Gap Bound

The coupling coefficient gap is bounded by:

$$\sigma_{\mathrm{ctrl}}(x) = \beta\sum_j \left|\frac{\partial(L_gL_{\hat{f}}^{m-1}h)}{\partial x_j}\right| \sigma_{\mathrm{GP},j}(x)\cdot u_{\max} \tag{2}$$

where $u_{\max} = \sup_{u\in\mathcal{U}}\|u\|$. On the CCS, $\sigma_{\mathrm{ctrl}}/\sigma_{\mathrm{total}} < 0.15$ for both constraints.

1.11. Online GP Structural Degradation Envelope

Remark 1 (Structural envelope for online GP)**.** *When the GP is updated online, the i.i.d. assumption is violated. Assuming Lipschitz continuity of the posterior with respect to hyperparameters and an online confidence sequence framework [**?**], the per-step margin obeys* $\epsilon_{k+1}(x) \leq \epsilon_k(x) + C_\sigma\|\theta_{k+1}^{\mathrm{GP}} - \theta_k^{\mathrm{GP}}\| + C_\beta\sqrt{\ln(k+1)/(k+1)}$. *Constants* C_σ, C_β *are not operationally computed; the practical safeguard is the* ϵ*-floor regularization.*

References

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